

Project Title

Exploratory Market & Pricing Analysis of Car Rental Platforms

Problem Statement

- Which car makes and models are most popular in different cities?
- What is the typical rental price across major cities?
- Are some cities oversaturated while others show market gaps?
- Are ratings genuinely informative or suspiciously inflated?
- Can we predict rental price or demand using vehicle & location features?

Data Overview & Data Types

The dataset contains **5,851 rows** and **15 columns**, describing car rental listings including vehicle characteristics, pricing, ratings, and location information.

Column	Data Type	Description
<code>fuelType</code>	Object	Fuel type of the vehicle (Electric, Gasoline, Hybrid)
<code>rating</code>	Float64	Average user rating (1–5)
<code>renterTripsTaken</code>	Int64	Number of trips completed by the renter
<code>reviewCount</code>	Int64	Number of reviews for the vehicle
<code>location.city</code>	Object	City of the listing
<code>location.country</code>	Object	Country of the listing
<code>location.latitude</code>	Float64	Latitude coordinate
<code>location.longitude</code>	Float64	Longitude coordinate
<code>location.state</code>	Object	State abbreviation
<code>owner.id</code>	Int64	Unique identifier for the vehicle owner
<code>rate.daily</code>	Int64	Daily rental price (USD)
<code>vehicle.make</code>	Object	Vehicle manufacturer
<code>vehicle.model</code>	Object	Vehicle model name
<code>vehicle.type</code>	Object	Type of vehicle (SUV, Car, etc.)
<code>vehicle.year</code>	Int64	Manufacturing year of the vehicle

Dataset Understanding

Category	Columns
Vehicle	<code>fuelType</code> , <code>vehicle.make</code> , <code>vehicle.model</code> , <code>vehicle.type</code> , <code>vehicle.year</code>
Pricing	<code>rate.daily</code>
Demand	<code>RenterTripsTaken</code> , <code>reviewCount</code>
Ratings	<code>rating</code>
Location	<code>location.city</code> , <code>state</code> , <code>country</code> , <code>latitude</code> , <code>longitude</code>
Owner	<code>owner.id</code>

Dataset Health Check

Column	Missing	% Missing	Action
<code>fuelType</code>	75	1.3%	Impute (mode)
<code>rating</code>	501	8.6%	Careful handling
Others	0	0%	Clean

Data Cleaning & Missing Value Treatment

Overview

The dataset contained missing values in selected categorical and numerical fields. A structured imputation strategy was applied to ensure data completeness while preserving business relevance.

Handling Missing Values in `fuelType`

Problem

- **Missing values:** 75
- **Percentage:** ~1.3% of total records

Given the small proportion of missing values and the strong dominance of certain fuel types, a simple but effective strategy was adopted.

Approach

Missing values in `fuelType` were imputed using the **mode (most frequent category)**.

Justification

- Minimal data loss

- Preserves categorical distribution
- Suitable for low-percentage missing categorical data

Handling Missing Values in rating

Problem

- **Missing values:** 501
- **Percentage:** ~8.6% of total records

Exploratory analysis revealed that missing ratings were primarily associated with listings having **zero reviews**, indicating unrated vehicles rather than data quality issues.

Approach

A **two-step hierarchical imputation strategy** was used:

1. **City-wise Median Imputation**
missing ratings were filled using the median rating of the respective city to preserve local rating behavior.
2. **Global Median Fallback**
For cities where no ratings were available, remaining null values were filled using the overall dataset median.

Justification

- Preserves geographical rating patterns
- Avoids bias from global mean imputation
- Improves robustness of downstream modeling
- Retains original rating availability information

Research Questions & Analytical Approaches

Which car makes and models are most popular in different cities?

Approach

- Grouped the dataset by `location.city`, `vehicle.make`, and `vehicle.model`.
- Used `renterTripsTaken` as a proxy for demand.

- Calculated average trips per make–model combination to identify popularity across cities.
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What is the typical rental price across major cities?

Approach

- Aggregated data at the city level.
 - Computed the mean of `rate.daily` for each city.
 - Compared average rental prices across major cities using summary statistics and visualizations.
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Are some cities oversaturated while others show market gaps?

Approach

- Measured supply using the total number of listings per city.
 - Measured demand using average `renterTripsTaken` per city.
 - Compared city-level supply and demand against overall averages.
 - Classified cities into oversaturated, market gap, or balanced categories.
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Are ratings genuinely informative or suspiciously inflated?

Approach

- Analyzed the distribution of `rating` values.
- Examined the relationship between `rating` and `reviewCount`.
- Identified cases with high ratings but low review counts to assess potential inflation.

Exploratory Market & Pricing Analysis – Results

1. Most Popular Car Makes & Models by City

Demand analysis using average renterTripsTaken shows that compact and economical models dominate city-level demand. Examples include Scion xB in Portland, FIAT 500 in New Orleans, Chevrolet Cruze in Irving, Mazda 6 in Miami, and BMW Z4 in Millbrae. These models indicate strong preference for affordable or fuel-efficient vehicles.

2. Typical Rental Price Across Major Cities

City-level average daily rental price analysis shows significant variation. Premium locations such as Tyson's Corner, Monument, and Beverly Hills exhibit higher average prices, while smaller or less tourist-heavy cities show moderate pricing. This reflects demand intensity, tourism, and income levels.

3. Market Saturation vs Market Gaps

Cities were classified based on supply (total cars) and demand (average trips). High-potential cities like Addison and Aberdeen Township show strong demand with relatively fewer listings, indicating market gaps. Oversaturated cities exhibit high supply but low demand, suggesting competition pressure.

4. Rating Reliability Analysis

Rating distribution analysis shows heavy concentration between 4.7 and 5.0. While ratings are generally positive, the limited variance and clustering at the top indicate possible rating inflation. Ratings alone may not fully reflect vehicle quality without considering review count.

5. Price & Demand Prediction Using Vehicle and Location Features

A regression model using vehicle characteristics and location features achieved a very high R^2 score (~ 0.99), indicating strong predictive power. Vehicle type, location, and model significantly influence rental price and demand, confirming that data-driven pricing strategies are feasible.